

Education

Bachelor of Technology (B.Tech) in Computer Science and Engineering (CSE)

Vellore Institute of Technology, Chennai

Expected Graduation: 2026

Experience

Research Intern

May 2025 – July 2025

Centre for e-Automation Technologies (CeAT), VIT Chennai

- Developed a **CloudSim Plus** framework to evaluate the **Hippopotamus Optimization (HO)** algorithm for **VM placement** in cloud data centers.
- Enhanced simulation scalability by modularizing components, enabling reproducible testing across multiple datacenter configurations — GitHub: Repo

Full Stack Developer Intern

Dec 2024 – Feb 2025

Daira Edtech Pvt Limited

- Architected the Activity Management System** and Student Progress modules using **Node.js** and **MongoDB**, enabling Faculty to manage project lifecycles and Students to track academic milestones and research paper submissions.
- Engineered Role-Based Access Control (RBAC)** for external stakeholders (Industry Experts & Reviewers) by implementing secure **JWT authentication** flows and custom RESTful API endpoints for profile management.
- Improved codebase maintainability** by refactoring legacy server logic, implementing rigorous server-side testing, and leading the debugging process for API errors to reduce system downtime.

Web Development Intern

Sept 2024 - Oct 2024

Indian Institute of Technology Bombay (IIT Bombay)

- Reduced data payload size by 90% for low-bandwidth environments by replacing text messages with numerical status codes in JSON responses.
- Enhanced backend robustness and API reliability by implementing Joi schema validation for the SaaS platform.

Projects

Water Brakes

Aug 2024 - Nov 2024

- Developed a Streamlit app that analyzes contour map data to recommend precise swale and trench locations for water management.
- Enabled data-driven groundwater recharge decisions by processing UTM data to generate interactive Plotly visualizations.

Technologies Used: Streamlit, Python, Plotly, Matplotlib | Live: [water-brakes.streamlit.app](#)

High-Performance Parallel AES Encryption Framework

Jan 2025 - Apr 2025

- Reduced encryption time by 51% on large payloads by engineering a parallelized AES system using Python's ProcessPoolExecutor.
- Increased throughput by 21.9% (to 362 KB/sec) compared to standard AES by designing a multi-core CTR-mode encryption pipeline.
- Achieved a 49.95% avalanche effect, confirming cryptographic robustness and validating the dynamic S-box security approach.

Technologies Used: Python, Multiprocessing, SHA-512, Matplotlib, NumPy

Hybrid Ensemble Learning for Early Abiotic Stress Detection in Crops

Aug 2025 – Nov 2025

- Developed a hybrid ensemble pipeline combining an LSTM with additive attention and XGBoost for early-stage abiotic stress detection using 30-day physiological sensor time-series data.
- Achieved **99.78% accuracy** and a **0.954 Macro F1 score** by learning robust temporal representations that capture dynamic plant stress patterns.
- Improved performance on limited datasets using an **8-timestep sliding window**, generating **64-dimensional temporal embeddings** that enhanced model interpretability and early detection reliability.

Technologies: Python, LSTM, Attention Mechanisms, XGBoost, Scikit-learn, Pandas, Time-Series Analysis

Skills

- Programming Languages:** Python, C++, C, Java, HTML, CSS, SQL, JavaScript, Bash.
- Databases & Cloud:** MongoDB, PostgreSQL, Amazon Web Services (AWS), Google Cloud (GCP), Docker.
- Tools & Frameworks:** Git, React, Next.js, Node.js, Express.js, RESTful APIs, Linux.
- Core Concepts:** Data Structures & Algorithms, DBMS, CN, OS, Cryptography (SHA-512/AES), CI/CD.

Positions of Responsibility

• Outreach Lead, DAO Community:

- Grew community membership by **50% in 6 months** through structured outreach and on-campus workshops in collaboration with Web3 companies including **Polygon, Spheron, Huddle01, blockOn, and wazirX**.
- Secured partnerships for **15+ national hackathons** and **TOKEN2049 Singapore**; led organization of **5 large-scale hackathons** ($\approx$ 250 participants each) and managed a **35-member team** for Defy24, achieving record sponsorship and profit.

• Blockchain Lead, Hackclub:

- Scaled the blockchain department by **300% in 6 months**, recruiting and mentoring **25+ junior developers**.
- Designed and delivered blockchain workshops, learning roadmaps, and personalized mentorship to upskill members.